Spencer Hilligoss

Irvine, CA   shilligo@uci.edu   (+1) 704-776-1680   LinkedIn

Education

PhD in Statistics *Sep 2023 – Present*
University of California, Irvine *(Expected June 2027)*

- **Coursework:** Advanced Probability, Machine Learning, Statistical Consulting
- **Research Topics**: Time-series Forecasting, Deep Learning, Generative Modeling, Causal Inference
- **Supervised by:** Dr. Annie Qu and Dr. Tianchen Qian
- **GPA:** 3.929/4.0

B.S. in Statistics and Operations Research, B.A. in Economics *Aug 2019 – May 2023*
University of North Carolina at Chapel Hill

- **Coursework:** Probability, Statistical Machine Learning, Real Analysis, Stochastic Modeling, Dynamic Decision Making, Financial Markets, Micro and Macro Economic Theory
- **GPA:** 3.7/4.0

Research and Work Experience

The Lasting Impact of HOLC Redlining on Minorities in New York *June 2022 - Aug 2022*
Harvard University, Research and Policy Institute

- Built a reproducible big-data workflow in R to merge HOLC redlining maps with demographic and socioeconomic datasets and construct neighborhood-level exposure measures.
- Estimated regression-based effects of historical redlining on present-day minority outcomes using large-scale, geographically linked data; ran robustness/sensitivity checks.

Sentiment-Enhanced LSTM Forecasting (DJIA) *Sep 2024 - Dec 2024*
University of California, Irvine

- Engineered a daily feature set combining market data with text-based sentiment signals (news/social text) and aligned features to next-day index returns.
- Implemented and trained LSTM forecasting models; compared against baseline time-series models using rolling/expanding-window backtests and out-of-sample error metrics.

Causal Mediation Pathways in Continuous Postprandial Glucose Monitoring for Type 1 Diabetes Patients *Sep 2024 – July 2026*
University of California, Irvine

- Accepted for publication at *npj Metabolic Health and Disease*.
- Developed a causally-constrained linear autoencoder (CLAE) to learn low-dimensional representations of high-dimensional longitudinal time-series data while preserving time-ordered causal structure.
- Built an end-to-end causal mediation workflow on the learned representations to estimate time-varying direct and mediated effects over 36 sequential time steps.

Selective Inference for Time-Varying Causal Effect Moderation via the Ordered Lasso *(Manuscript in Preparation)* *Feb 2026 – Present*
University of California, Irvine

- Developing selective inference procedures to identify time-varying moderators of causal effects using the ordered lasso, with applications to mobile health.

Fellowships and Awards

University of California, Irvine

- T32 STEER in Biomedical Sciences Fellow *Aug 2025 - Present*
- Diversity Recruitment Fellowship *Aug 2023*

Mentorship

Undergraduate Research Mentor, University of California, Santa Barbara

- Bryson Harris — financial time series modeling (weekly research meetings).
- Daniel You — generative diffusion modeling with an application to soccer tactics (weekly research meetings).

Computing Tools

Python (PyTorch, NumPy, Pandas, scikit-learn); R; Git; Linux; HPC/Slurm; Stata; Excel